

GRADE 9 MATHEMATICS · NUMBER

Square Root & Radicals

THE BIG IDEA

√ always means the **positive**
Pythagorean theorem only

PYTHAGOREAN THEOREM

$$a^2 + b^2 = c^2$$

c = hypotenuse (opposite the right angle, always

S

**ve root — and the
works on right triangles.**

s the longest side)

$$\sqrt{(a \times b)} = \sqrt{a} \times \sqrt{b} \quad \checkmark \quad \sqrt{(a + b)} \neq \sqrt{a} + \sqrt{b} \quad \times$$

WORKED EXAMPLE

Right triangle: legs 7 cm and 24 cm

$$7^2 + 24^2 = c^2$$

$$49 + 576 = 625$$

$$c = \sqrt{625} = 25 \text{ cm}$$

⚠ WATCH OUT FOR

MISTAKE

$$\sqrt{9} = \pm 3$$

MISTAKE

$$\sqrt{(9 + 16)} = 3 + 4 = 7$$

FIX

The principal root is positive only: **3**

FIX

Roots don't distribute over addition: $\sqrt{25} = \mathbf{5}$

MISTAKE

" $\sqrt{2} = 1.414...$ exactly"



SELF-CHECK

Can you estimate $\sqrt{70}$ without a calculator? Can you draw a right triangle?

That's an approximation — the exact value is $\sqrt{2}$

Can you find a missing side in a right